Based on your **Jupyter Notebook** and the **Power BI Risk Analysis Dashboard**, here is a professional business report with actionable insights, business problems, and analytical questions suitable for a portfolio project.

Banking Customer Risk Analysis Report

Executive Summary

The analysis covers **3,000 banking customers** to understand customer risk, income segments, loyalty behavior, and financial exposure. The objective is to identify high-risk customers, reduce credit losses, and support better lending decisions.

Dashboard KPIs

KPI	Value	Business Meaning
Total Customers	3,000	Customer base analyzed
Average Income	$171.31K	Strong earning customer base
Total Bank Deposits	$2.01B	Large deposit portfolio
Total Bank Loans	$1.77B	Significant lending exposure
Total Business Lending	$2.60B	High commercial lending portfolio

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Key Findings

1. Risk Distribution

- **Risk Level 2** is the largest customer segment.
- **Risk Level 5** has the fewest customers.
- Most customers belong to **low-to-medium risk categories**, indicating a relatively healthy customer portfolio.

Business Impact

- Current lending strategy appears effective.
- Small group of high-risk customers should be closely monitored.

Recommended Actions

- Continue approving loans for Risk 1–2 customers.
- Increase monitoring for Risk 4–5 customers.
- Review credit limits for high-risk accounts.

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2. Income Band Analysis

Distribution shows:

- Medium Income → Largest customer group
- Low Income → Second largest
- High Income → Smallest segment

Business Impact

The bank primarily serves middle-income customers.

Recommended Actions

- Introduce premium investment products for high-income customers.
- Offer salary accounts and personal loans to medium-income customers.
- Create financial literacy and savings programs for low-income customers.

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3. Nationality Distribution

European customers form the largest customer base, followed by Asian customers.

Business Impact

The customer portfolio is concentrated within a few nationalities.

Recommended Actions

- Expand marketing toward underrepresented nationalities.
- Create region-specific financial products.
- Analyze whether default patterns differ by nationality.

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4. Loyalty Classification

Customer distribution:

- Jade → Largest
- Silver → Moderate
- Gold → Smaller
- Platinum → Smallest

Notebook analysis also indicates:

- **Platinum customers have the safest risk profile.**

Business Impact

High loyalty is associated with lower banking risk.

Recommended Actions

- Upgrade eligible Jade customers to higher loyalty tiers.
- Provide rewards encouraging movement to Gold and Platinum.
- Use loyalty level in credit approval models.

Notebook Insights

Income vs Risk

The notebook identifies this as the strongest relationship.

Customers with different income bands show distinct risk patterns.

Business Action

Include income band as a major variable in credit scoring.

Loans vs Risk

Higher-risk customers generally require closer loan monitoring.

Business Action

- Tighten loan approval criteria.
- Increase repayment monitoring.
- Adjust interest rates based on risk.

Deposits vs Risk

Customers maintaining higher deposits generally present stronger financial stability.

Business Action

- Offer relationship banking.
- Encourage fixed deposits.
- Use deposit history when evaluating loans.

Correlation Analysis

Strong relationships found:

- Bank Deposits ↔ Checking Accounts
- Bank Deposits ↔ Savings Accounts
- Business Lending ↔ Bank Deposits

Business Action

Cross-sell banking products instead of marketing them independently.

Business Problems Identified

Problem 1

High-risk customers may increase loan default rates.

Solution
Implement automated early-warning alerts for Risk 4–5 customers.

Problem 2

Customer base is concentrated in medium-income segments.

Solution
Acquire more high-income customers through premium banking products.

Problem 3

Large number of Jade customers.

Solution
Increase loyalty upgrade campaigns to improve customer retention.

Problem 4

Business lending exposure is very high.

Solution
Review commercial borrowers periodically and reassess risk ratings.

Problem 5

High-risk customers may receive the same products as low-risk customers.

Solution
Adopt risk-based pricing and personalized lending policies.

Actionable Recommendations

Priority	Recommendation	Expected Benefit
High	Monitor Risk 4–5 customers monthly	Reduce defaults
High	Use income band in credit approval	Better risk prediction
High	Reward Platinum and Gold customers	Increase retention
Medium	Upsell investment products to high-incomecustomers	Increase revenue
Medium	Cross-sell savings and checking accounts	Increase deposits
Medium	Expand into underrepresented nationalities	Diversify customer portfolio
Medium	Review business lending periodically	Lower commercial credit risk

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Business Questions Answered

- Which customers are most likely to default?
- Which income group carries the highest banking risk?
- Does customer loyalty reduce financial risk?
- Which nationality contributes the most customers?
- Is the bank's lending portfolio concentrated in risky customers?
- Which customer segment should receive premium banking products?
- Which customers require closer monitoring?
- Which factors should influence future credit approval decisions?

Overall Conclusion

The bank maintains a **predominantly low-risk portfolio**, with **Risk Level 2 customers forming the largest segment**. The strongest insights from the notebook indicate that **income**

band and loyalty classification are closely related to customer risk, making them valuable predictors for lending decisions. By focusing on proactive monitoring of high-risk customers, expanding premium offerings to high-income clients, and encouraging loyalty upgrades, the bank can reduce credit risk while improving customer retention and profitability.